

Cultural Homophily in Grand Slam Doubles

ATP Men's Grand Slam Doubles, 2018–2025 · 1,793 matches after dropping 47 retirements / walkovers · Regression sample: 3,222 team-obs (complete ranking, nationality/language, and GS experience for all 4 players)

1. Data and Measures

Three culture measures capture teammate similarity. **Same nationality** and **same official language** are binary (0/1). **Language proximity** is a continuous 0–1 index of ethnolinguistic closeness (higher = more similar language family).

Regressions use a team-level panel with tournament-by-year fixed effects, round fixed effects, and standard errors clustered by match. Controls in all main specifications: team average doubles ranking, opponent average ranking, top-100 singles indicator (either player), team average GS doubles appearances (*exp_mean*), and its square. Culture measures enter one at a time. Sensitivity tables (3b, 4b, 5b) additionally include teammate rank gap to verify stability; rank gap is insignificant throughout and is excluded from main specifications. Nationality and language data are fully resolved in the pipeline (birthplace fallback, surname lookup, Monaco language correction, and `manual_nationality.csv`); zero matches are dropped for nationality.

2. Observation Breakdown

Step	Description	Matches	Change	Team-obs
1	Raw scraped dataset (2018–2025, excl. 2020)	1,840	—	3,680
2	Drop retirements / walkovers (20 walkovers · 9 retired in set 1 · 18 retired in set 2+)	1,793	–47	3,586
3	Drop: doubles ranking incomplete for ≥1 player	1,779	–14	3,558
4	Drop: nationality / language missing for ≥1 player (pipeline fully resolves: birthplace, surname lookup, Monaco fix, <code>manual_nationality.csv</code>)	1,779	0	3,558
5	Exclude Olympic matches (Tokyo 2021 · Paris 2024)	1,716	–63	3,432
6	Drop: GS experience missing for ≥1 player	—	–210 obs	3,222
→ Grand Slams regression sample (Tables 3–5)		—		3,222

Regression sub-samples: Table 4 (any tiebreak) — 1,596 obs (857 matches). Table 4d (7-pt tiebreaks only) — 1,435 obs (770 matches). Table 5 (adversity: lost set 1) — 1,592 obs (one per match). Table 5c (contested only: 0|1|1 + 0|1|0) — 710 obs.

3. Pressure Outcome Counts

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	445	24.8%
Set-2 tiebreak (7-pt)	470	26.2%
Any regular tiebreak (sets 1–2)	804	44.8%
Match tiebreak / super-tb (10-pt, set 3)	24	1.3%
Any tiebreak (all types)	896	50.0%
Match went to 3 sets	787	43.9%
Comeback wins (winner lost set 1)	351	19.6%

All matches (GS + Olympics, N=1,793). Match tiebreaks concentrated in Olympics (N=20) and Wimbledon post-2018 (N=4).

4. Olympic Cycle Descriptive Evidence

Table 2A — Tokyo 2021 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Tokyo	AO18 – USO19	1,006	40.1%	59.9%	55.8%	44.2%	56.7%	43.3%
Tokyo Prep	AO20 – Wim21 (ex Wim20)	364	42.0%	58.0%	57.0%	43.0%	57.0%	43.0%
Post-Tokyo	USO21 – USO22	604	39.2%	60.8%	55.4%	44.6%	56.3%	43.7%

N teams = GS matches × 2. Same/Diff = share of all teams (pooled winner + loser) on each cultural dimension.

Table 2B — Paris 2024 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Paris	AO22 – USO22	486	39.5%	60.5%	54.8%	45.2%	55.9%	44.1%
Paris Prep	AO23 – Wim24	860	42.6%	57.4%	58.5%	41.5%	58.9%	41.1%
Post-Paris	USO24 – USO25	626	43.9%	56.1%	58.9%	41.1%	60.4%	39.6%

N teams = GS matches × 2.

Cultural Homophily in Grand Slam Doubles - Tokyo 2021 Cycle
(Same vs. Diff, by Tournament-Year)

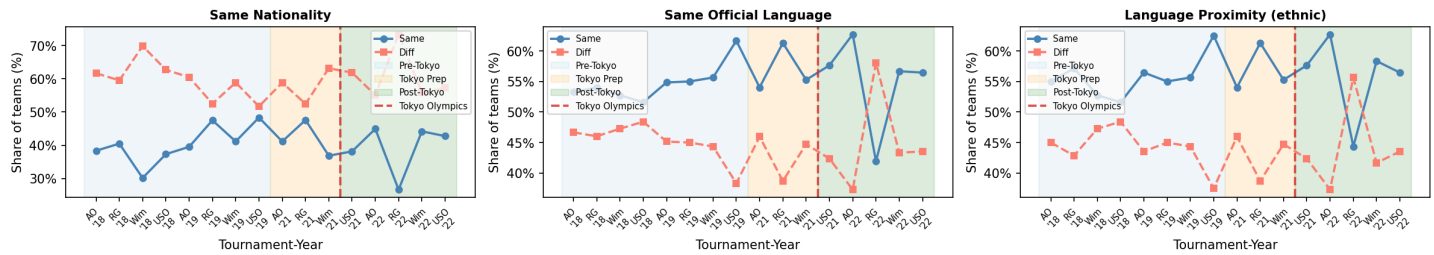


Figure 1. Share of same- vs. different-culture teams by tournament-year, Grand Slams only. Shaded: Pre-Tokyo (blue), Tokyo Prep (orange), Post-Tokyo (green). Red dashed = Tokyo Olympics 2021.

Cultural Homophily in Grand Slam Doubles - Paris 2024 Cycle
(Same vs. Diff, by Tournament-Year)

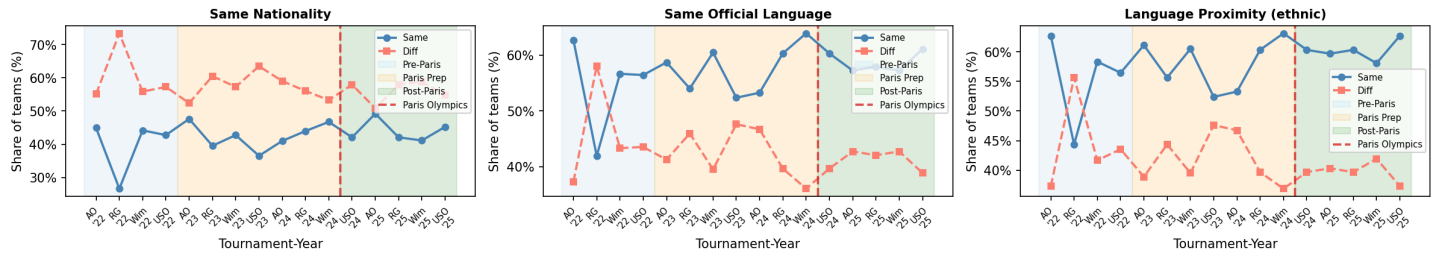


Figure 2. Paris 2024 cycle. Red dashed = Paris Olympics 2024.

Cultural Homophily in Grand Slam Doubles -- Full Timeline 2018-2025
(Tokyo & Paris Olympic Cycles Combined, Same vs. Diff)

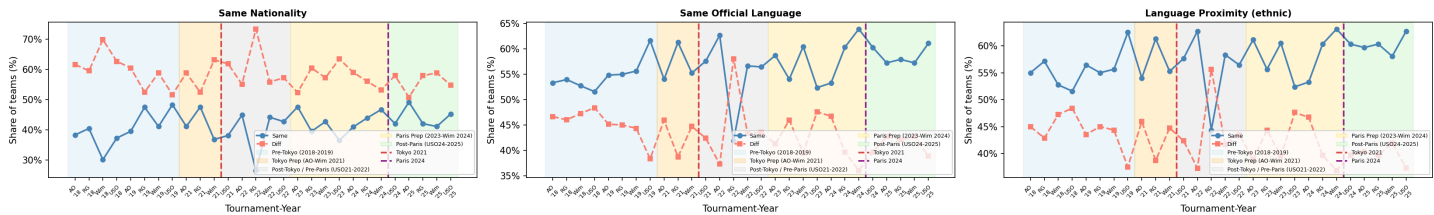


Figure 3. Both Olympic cycles on a single timeline (2018–2025). Vertical dashed lines mark Tokyo 2021 and Paris 2024.

5. Regression Results

Logit coefficients (log-odds). Each column is a separate model. Standard errors in parentheses, clustered by match (Tables 3–4); HC3 robust (Table 5, one obs per match). FE: tournament $\times$ year, round. All main specifications include GS experience controls (*exp_mean* and its square). Sensitivity tables (3b, 4b, 5b) also include teammate rank gap (consistently insignificant).

Table 3 — Outcome: Match Win | N = 3,222 team-obs

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1596** (0.0808) p=0.048	+0.2060*** (0.0776) p=0.008	+0.1885** (0.0775) p=0.015
Team avg. doubles rank	−0.0034*** (0.0004)	−0.0033*** (0.0004)	−0.0033*** (0.0004)
Opponent avg. doubles rank	+0.0033*** (0.0003)	+0.0033*** (0.0003)	+0.0033*** (0.0003)
Top-100 singles player	−0.0048 (0.1507)	+0.0022 (0.1508)	+0.0048 (0.1507)
Team avg GS appearances	+0.1039** (0.0416)	+0.1050** (0.0414)	+0.1030** (0.0414)
GS experience squared	−0.0033** (0.0015)	−0.0034** (0.0015)	−0.0033** (0.0015)
Pseudo-R ²	0.096	0.097	0.097

*** p<0.01 ** p<0.05 * p<0.10. Clustered SE. FE: tournament×year, round. All three culture measures statistically significant (same language p=0.008, now p<0.01). Experience shows a concave effect: positive linear (p≈0.01), negative quadratic (p≈0.03).

Table 3b — Sensitivity: Match Win + Teammate Rank Gap | N = 3,222 team-obs

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1578 [*] (0.0808) p=0.051	+0.2031 ^{***} (0.0776) p=0.009	+0.1851 ^{**} (0.0775) p=0.017
Team avg. doubles rank	−0.0040 ^{***} (0.0007)	−0.0040 ^{***} (0.0007)	−0.0040 ^{***} (0.0007)
Opponent avg. doubles rank	+0.0032 ^{***} (0.0003)	+0.0032 ^{***} (0.0003)	+0.0032 ^{***} (0.0003)
Top-100 singles player	+0.0296 (0.1539)	+0.0343 (0.1541)	+0.0371 (0.1539)
Team avg GS appearances	+0.0946 ^{**} (0.0423)	+0.0956 ^{**} (0.0420)	+0.0937 ^{**} (0.0420)
GS experience squared	−0.0030 ^{**} (0.0015)	−0.0031 ^{**} (0.0015)	−0.0030 ^{**} (0.0015)
Teammate rank gap	+0.0008 (0.0005)	+0.0008 (0.0005)	+0.0008 (0.0005)
Pseudo-R ²	0.092	0.093	0.093

Culture coefficients essentially unchanged from Table 3. Rank gap insignificant (p≈0.15–0.16).

Table 4 — Outcome: Match Win (any tiebreak) | N = 1,596 team-obs (857 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1949 [*] (0.1156) p=0.092	+0.2082 [*] (0.1090) p=0.056	+0.1979 [*] (0.1087) p=0.069
Team avg. doubles rank	−0.0025 ^{***} (0.0005)	−0.0025 ^{***} (0.0005)	−0.0025 ^{***} (0.0005)
Opponent avg. doubles rank	+0.0021 ^{***} (0.0004)	+0.0020 ^{***} (0.0004)	+0.0021 ^{***} (0.0004)
Top-100 singles player	+0.3555 (0.2226)	+0.3626 (0.2228)	+0.3646 (0.2226)
Team avg GS appearances	+0.0645 (0.0573)	+0.0648 (0.0571)	+0.0631 (0.0571)
GS experience squared	−0.0017 (0.0021)	−0.0017 (0.0021)	−0.0016 (0.0021)
Pseudo-R ²	0.042	0.042	0.042

Sample: GS matches with any tiebreak (7-pt sets 1–2 or 10-pt match TB). All three culture measures marginally positive (p=0.056–0.092). See Table 4d for 7-pt tiebreaks only.

Table 4b — Sensitivity: Tiebreak Sample + Rank Gap | N = 1,596 team-obs

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1950 [*] (0.1157) p=0.092	+0.2083 [*] (0.1091) p=0.056	+0.1980 [*] (0.1088) p=0.069
Team avg. doubles rank	−0.0025 ^{***} (0.0008)	−0.0024 ^{***} (0.0008)	−0.0024 ^{***} (0.0008)
Opponent avg. doubles rank	+0.0021 ^{***} (0.0004)	+0.0020 ^{***} (0.0004)	+0.0021 ^{***} (0.0004)
Top-100 singles player	+0.3539 (0.2218)	+0.3609 (0.2221)	+0.3630 (0.2219)
Team avg GS appearances	+0.0654 (0.0580)	+0.0657 (0.0577)	+0.0640 (0.0577)
GS experience squared	−0.0017 (0.0021)	−0.0017 (0.0021)	−0.0017 (0.0021)
Teammate rank gap	−0.0001 (0.0006)	−0.0001 (0.0006)	−0.0001 (0.0006)
Pseudo-R ²	0.042	0.042	0.042

Culture coefficients unchanged. Rank gap insignificant (p≈0.90).

Table 4c — Match Win, Last-Set Tiebreak Only | N = 371 team-obs (199 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1458	+0.0850	+0.0609
	(0.2600)	(0.2412)	(0.2405)
	p=0.575	p=0.724	p=0.800
Team avg. doubles rank	−0.0020 **	−0.0020 **	−0.0020 **
	(0.0010)	(0.0010)	(0.0010)
Opponent avg. doubles rank	+0.0012	+0.0011	+0.0011
	(0.0008)	(0.0008)	(0.0008)
Top-100 singles player	+1.2753 **	+1.2812 **	+1.2806 **
	(0.5581)	(0.5564)	(0.5559)
Team avg GS appearances	−0.1963	−0.2031	−0.2032
	(0.1244)	(0.1237)	(0.1236)
GS experience squared	+0.0087 *	+0.0090 *	+0.0090 *
	(0.0047)	(0.0046)	(0.0046)
Pseudo-R ²	0.039	0.038	0.038

Sample: matches decided by a last-set tiebreak (set-3 7-pt: 195 matches; 10-pt match TB: 4 matches). Culture coefficients insignificant in this sub-sample. Top-100 singles strongly significant (p≈0.03).

Table 4d — Match Win, 7-Point Tiebreaks Only | N = 1,435 team-obs (770 matches)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.1952 (0.1245) p=0.117	+0.1905 (0.1173) p=0.104	+0.1845 (0.1170) p=0.115
Team avg. doubles rank	−0.0028*** (0.0006)	−0.0028*** (0.0006)	−0.0027*** (0.0006)
Opponent avg. doubles rank	+0.0023*** (0.0005)	+0.0023*** (0.0005)	+0.0023*** (0.0005)
Top-100 singles player	+0.2337 (0.2376)	+0.2440 (0.2380)	+0.2457 (0.2379)
Team avg GS appearances	+0.0747 (0.0617)	+0.0748 (0.0614)	+0.0734 (0.0614)
GS experience squared	−0.0020 (0.0023)	−0.0021 (0.0023)	−0.0020 (0.0023)
Pseudo-R ²	0.049	0.049	0.049

Sample: GS matches with a regular 7-pt tiebreak in sets 1 or 2 (excludes 10-pt match TBs). Results consistent with Table 4: culture coefficients positive but individually insignificant. Restricting to homogeneous tiebreak type does not sharpen the pattern.

Table 5 — Outcome: Comeback Win (Adversity Sample) | N = 1,592 obs (1 per match)

Mean ranks by adversity group (1,674 GS matches, before experience filter)

Group	Description	N	Team rank	Opp. rank	Rank gap
o 1 1	Comeback win (lost S1, won S2 and S3)	334	101.5	120.4	65.8
o 1 0	Three-set loss (lost S1, won S2, lost S3)	428	147.1	74.2	101.8
o 0	Straight-set loss (lost S1 and S2)	954	202.9	82.6	149.3

Mean doubles ranking (lower = better). Comeback teams (o|1|1) are substantially stronger than both the three-set-loss group (102 vs. 144) and the swept group (102 vs. 201), indicating quality confounding. Table 5c compares the two more-balanced groups (o|1|1 vs. o|1|0) after removing the o|0 group.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	−0.1558 (0.1384) p=0.260	−0.2258* (0.1332) p=0.090	−0.2214* (0.1332) p=0.096
Team avg. doubles rank	−0.0024*** (0.0006)	−0.0024*** (0.0006)	−0.0024*** (0.0006)
Opponent avg. doubles rank	+0.0019*** (0.0004)	+0.0019*** (0.0004)	+0.0019*** (0.0004)
Top-100 singles player	−0.2799 (0.2589)	−0.2825 (0.2585)	−0.2791 (0.2561)
Team avg GS appearances	−0.0098 (0.0657)	−0.0102 (0.0654)	−0.0053 (0.0650)
GS experience squared	+0.0005 (0.0024)	+0.0005 (0.0024)	+0.0003 (0.0024)
Pseudo-R ²	0.059	0.060	0.056

*** p<0.01 ** p<0.05 * p<0.10. HC3 robust SE (one obs per match). FE: tournament×year, round (col. 3: tournament×year only due to near-collinearity). N=1,592 after experience filter. Outcome: 0|1|1 (19.5%), 0|1|0 (24.9%), 0|0 (55.6%). Same language and proximity marginally negative (p≈0.09–0.10). Same nationality not significant (p=0.26).

Table 5c — Comeback Win, Contested Matches Only (0|1|1 + 0|1|0) | N = 710 obs

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	-0.3376 [*] (0.1786) p=0.059	-0.3477 ^{**} (0.1717) p=0.043	-0.3065 [*] (0.1688) p=0.070
Team avg. doubles rank	-0.0016 ^{***} (0.0006)	-0.0016 ^{***} (0.0006)	-0.0016 ^{***} (0.0006)
Opponent avg. doubles rank	+0.0023 ^{***} (0.0008)	+0.0023 ^{***} (0.0007)	+0.0023 ^{***} (0.0007)
Top-100 singles player	+0.1051 (0.3451)	+0.0795 (0.3439)	+0.0774 (0.3440)
Team avg GS appearances	+0.0035 (0.0809)	+0.0134 (0.0803)	+0.0153 (0.0802)
GS experience squared	-0.0004 (0.0030)	-0.0008 (0.0030)	-0.0009 (0.0030)
Pseudo-R ²	0.123	0.124	0.124

HC3 robust SE. FE: tournament×year, round. Sample: teams that lost set 1 and went to three sets (0|1|1 + 0|1|0); straight-set losses (0|0) excluded. Mean ranks: 0|1|1 = 102, 0|1|0 = 144 — more balanced than the full adversity sample. N = 710 after experience filter; comeback share = 43.8%. Culture measures negative: same language significant (p=0.043), nationality and proximity marginal (p≈0.06–0.07). Same-culture teams are less likely to mount a comeback in contested three-set matches. Col. (3) uses tournament×year FE only due to near-singular matrix with round FE.

Table 5b — Sensitivity: Adversity Sample + Teammate Rank Gap | N = 1,592 obs

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	−0.1551	−0.2259*	−0.2220*
	(0.1384) p=0.263	(0.1331) p=0.090	(0.1332) p=0.096
Team avg. doubles rank	−0.0026**	−0.0026**	−0.0026**
	(0.0010)	(0.0011)	(0.0010)
Opponent avg. doubles rank	+0.0018***	+0.0019***	+0.0018***
	(0.0004)	(0.0004)	(0.0004)
Top-100 singles player	−0.2765	−0.2784	−0.2827
	(0.2598)	(0.2595)	(0.2594)
Team avg GS appearances	−0.0117	−0.0123	−0.0107
	(0.0658)	(0.0656)	(0.0655)
GS experience squared	+0.0006	+0.0006	+0.0005
	(0.0024)	(0.0024)	(0.0024)
Teammate rank gap	+0.0002	+0.0002	+0.0002
	(0.0008)	(0.0008)	(0.0008)
Pseudo-R ²	0.059	0.060	0.060

HC3 robust SE. Culture coefficients essentially unchanged from Table 5. Rank gap insignificant (p≈0.80).